

### Lab 3:

Perform the following DB operations using Cassandra.

- Create a keyspace by name Library
- Create column family Library\_Info: Stud\_Id (PK), Counter\_value (Counter), Stud\_Name, Book\_Name, Book\_Id, Date\_of\_issue
- Insert values into the table in batch
- Display the table and increase the value of the counter
- Write a query to show student id 112 has taken book 'BDA' 2 times
- Export the created column to a csv file
- Import a given csv dataset into Cassandra column family

```
CREATE KEYSPACE IF NOT EXISTS Library
  WITH replication={'class':'SimpleStrategy','replication_factor':1};
USE Library;
```

```
CREATE TABLE IF NOT EXISTS Library_Info (
  Stud_Id      INT PRIMARY KEY,
  Stud_Name    TEXT,
  Book_Name    TEXT,
  Book_Id      TEXT,
  Date_of_issue DATE
);
```

```
CREATE TABLE IF NOT EXISTS Book_Counter (
  Stud_Id      INT,
  Book_Name    TEXT,
  Counter_value COUNTER,
  PRIMARY KEY((Stud_Id), Book_Name)
);
```

```
BEGIN BATCH
  INSERT INTO Library_Info(Stud_Id,Stud_Name,Book_Name,Book_Id,Date_of_issue)
    VALUES(112,'Tarun S','BDA','B101','2026-02-13');
  INSERT INTO Library_Info(Stud_Id,Stud_Name,Book_Name,Book_Id,Date_of_issue)
    VALUES(113,'Tanu R','AI','B102','2026-02-13');
APPLY BATCH;
```

```
UPDATE Book_Counter SET Counter_value=Counter_value+1
  WHERE Stud_Id=112 AND Book_Name='BDA';
UPDATE Book_Counter SET Counter_value=Counter_value+1
  WHERE Stud_Id=112 AND Book_Name='BDA';
```

```
SELECT * FROM Book_Counter WHERE Stud_Id=112 AND Book_Name='BDA';
```

```
COPY Library_Info TO '/home/bmscecse/Desktop/Library_Info.csv';
```

```
COPY Library_Info FROM '/home/bmscecse/Desktop/Library_Info.csv';
```

```
// e) Counter result:
stud_id | book_name | counter_value
-----+-----+-----
```

```
112 | BDA | 2  
(1 rows)
```

```
// f) Export:
```

```
Using 16 child processes
```

```
2026-02-13T11:14:55.192+0530 connected to: localhost/
```

```
2 rows exported to 1 files in 0.124 seconds.
```

```
// g) Import:
```

```
2026-02-13T11:15:05.360+0530 connected to: localhost/
```

```
2 rows imported from 1 files in 0.377 seconds (0 skipped).
```